Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ______________
Topic: Linear Inequalities with Two Variables **Date:** __________

Advanced (Habanero)

Q1. What is the graph of the inequality $y > 2x - 3$?
(A) Solid line with shading above
(B) Dashed line with shading below
(C) Solid line with shading below
(D) Dashed line with shading above
(E) No shading

Q2. The inequality $x + 2yleq4$ has a
(A) Solid boundary line
(B) Dashed boundary line
(C) No boundary line
(D) Vertical boundary line
(E) Horizontal boundary line

Q3. Which inequality represents the region below the line $y = 3x - 2$?
(A) $yleq3x - 2$
(B) $ygeq3x - 2$
(C) $y < 3x - 2$
(D) $y > 3x - 2$
(E) $y = 3x - 2$

Q4. The inequality $y > -2x + 1$ has a
(A) Dashed boundary line
(B) Solid boundary line
(C) No boundary line
(D) Vertical boundary line
(E) Horizontal boundary line

Q5. What is the solution region of the inequality $x - 2y > 3$?
(A) Above the line $x - 2y = 3$
(B) Below the line $x - 2y = 3$
(C) On the line $x - 2y = 3$
(D) To the left of the line $x - 2y = 3$
(E) To the right of the line $x - 2y = 3$

Q6. The inequality $2x + 5ygeq10$ represents the region
(A) Above the line $2x + 5y = 10$
(B) Below the line $2x + 5y = 10$
(C) On the line $2x + 5y = 10$
(D) To the left of the line $2x + 5y = 10$
(E) To the right of the line $2x + 5y = 10$

Q7. Which inequality has the solution region below the line $y = -x + 2$?
(A) $yleq - x + 2$
(B) $ygeq - x + 2$
(C) $y < -x + 2$
(D) $y > -x + 2$
(E) $y = -x + 2$

Q8. The inequality $y < 4x - 1$ represents the region
(A) Below the line $y = 4x - 1$
(B) Above the line $y = 4x - 1$
(C) On the line $y = 4x - 1$
(D) To the left of the line $y = 4x - 1$
(E) To the right of the line $y = 4x - 1$

Open Ended Questions (FRQ)

Q9. Graph the inequality $x + y > 2$ and identify the boundary line.

Q10. Find the solution region of the system of inequalities $y > x - 1$ and $y < -x + 3$.

Chef's Tips

- Tip 1: Emphasize the importance of solid and dashed boundary lines.
- Tip 2: Focus on the direction of the inequality to determine the solution region.
- Tip 3: Use graphing to visualize the inequalities and systems of inequalities.